

# Short notes

Understand the idea. Remember the essentials. Check the source.

## 01 Electric charge

### 💡 Key concept

Electric charge is a property of matter that causes electrical interactions.

📖 p.1 S1

### 🎯 Check your understanding

What happens when two positive charges are brought near each other?

📖 p.1 S1

### Must-know points

- Two types: positive and negative.  
📖 p.1 S1
- Like charges repel; unlike charges attract.  
📖 p.1 S1
- SI unit: coulomb (C).  
📖 p.1 S1
- Conserved in an isolated system; quantised in multiples of  $e$ .  
📖 p.1 S1

📖 p.1 S1

## 02 Coulomb's law

### 💡 Key concept

Coulomb's law gives the electrostatic force between two stationary point charges.

📖 p. 2 S2

### 🎯 Check your understanding

What happens to force when separation doubles?

📖 p. 2 S2

### Must-know points

- $F = k |q_1 q_2| / r^2$ .
- Force acts along the line joining the charges.
- Double  $r \rightarrow$  force becomes  $F/4$ .
- Net force is the vector sum of individual forces.

📖 p. 2 S2

📖 p. 2 S2

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📖 p. 2 S2

### ⚗ Formula & units

$F = k |q_1 q_2| / r^2$ ;  $k = 1/(4\pi\epsilon_0)$  in vacuum.

📖 p. 2 S2

### 💡 Example from the evidence

If separation changes from  $r$  to  $2r$ , the force changes from  $F$  to  $F/4$ .

📖 p. 2 S2

📖 p. 2 S2

## 03 Electric field

### 💡 Key concept

Electric field is the force per unit positive test charge:

$$E = F/q.$$

📖 p. 3 S3

### 🎯 Check your understanding

Why should a test charge be small?

📖 p. 3 S3

### Must-know points

- Vector quantity: magnitude and direction.  
📖 p. 3 S3
- SI unit: N/C.  
📖 p. 3 S3
- Use a small test charge to avoid disturbing source charges.  
📖 p. 3 S3
- Field lines go away from positive charge and toward negative charge; they do not intersect.  
📖 p. 3 S3

### ⚗ Formula & units

$$E = F/q; \text{ unit} = \text{N/C}.$$

📖 p. 3 S3

📖 p. 3 S3

## 04 Electric potential

### 💡 Key concept

Electric potential is potential energy per unit charge:  $V = U/q$ .

📖 p. 4 S4

### ⚠️ Avoid this confusion

Incorrect: zero potential always means zero field. Correct: a field can exist where potential is zero.

📖 p. 4 S4

### Must-know points

- Scalar quantity: no vector direction.

📖 p. 4 S4

- SI unit: volt = joule/coulomb.

📖 p. 4 S4

- Potential difference is change in potential energy per unit charge between two points.

📖 p. 4 S4

- Zero potential does not necessarily mean zero electric field.

📖 p. 4 S4

### 🎯 Check your understanding

Is electric potential a vector?

📖 p. 4 S4

### ⚗️ Formula & units

$V = U/q$ ; unit = volt = J/C.

📖 p. 4 S4

📖 p. 4 S4